

SIMD-Optimized ISAP Algorithm for Enhanced Performance in Resource-Constrained Embedded Systems

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Abstract—This project achieves a low power embedded systems implementation of ISAP with low latency. The low latency is achieved by implementing two changes to the baseline design: a low latency version of ASCON’s permutation (10 rounds instead of 12) and the integration of a logic gate that creates a parallelizable approach inspired by SIMD. Thus, this implementation is low latency and low memory without compromising ISAP’s security expectations. Latency and memory benchmarks are provided for four different ISAP constructions (ISAP-A-128, ISAP-A-128A, ISAP-K-128, and ISAP-K-128A) across three different microcontrollers and construction settings (Arduino Uno, ESP32, Windows Subsystem for Linux). All versions in all environments show reduced time and memory allocations during execution. The construction is also remains secure against side-channel attacks while providing low latency in a low power, resource constrained setting with throughput improvements of 20-35%.

Index Terms—Lightweight cryptography, ISAP, SIMD optimization, embedded security, AEAD, side-channel resistance

I. INTRODUCTION

As IoT devices and embedded applications become more popular, the necessity for cryptography becomes lighter to accommodate resource scarce alternatives. Standard authenticated encryption will fail resource requirements for microcontrollers—authentication levels to be cryptographically secure result in too much overhead in computation and memory for low throughput low power capabilities [1].

Lightweight ISAP is a CAESAR AEAD candidate for authenticated encryption which provides additional side channel resilience via domain separation of key materials [2]. However, using a 12-round ASCON permutation in its finished form as a requirement may challenge performance in low-power applications.

This work features an ISAP implementation from the ground up with a requirement of a 10-round ASCON permutation, allowing for potential parallel processing methods to be leveraged via SIMD to reduce performance time and increase

throughput without jeopardizing authentication quality or side-channel resilience.

Four ISAP variants—ISAP-A-128, ISAP-A-128A, ISAP-K-128, and ISAP-K-128A—were tested on three different hardware platforms: Windows Subsystem for Linux (WSL), Arduino Uno, and ESP32. Our benchmarks demonstrate significant gains in throughput, memory efficiency, and encryption/decryption speed—all while upholding robust security guarantees. These findings show that SIMD-optimized ISAP can balance robustness and performance even in limited settings, making it a practical option for real-world embedded systems.

PROBLEM STATEMENT

Even though ISAP was made for authenticated encryption that resists side channels and works well in low-resource environments, it depends on a 12-round ASCON permutation that mainly introduces overhead for low-power devices like microcontrollers. This greatly hinders the practical deployment of ISAP in latency-critical, always-on battery-operated embedded systems. Furthermore, the original design seems not to exploit the instruction-level parallelism that various platforms afford, nor does it use any memory layout optimizations.

This work addresses the problem of how to substantially enhance the performance of ISAP on limited devices without degrading its fundamental security features. Our specific targets are execution time and memory usage—making improvements in these areas while still keeping ISAP safe against side-channel attacks and compatible with a range of embedded platforms.

II. RELATED WORK

A. Lightweight Cryptography Evolution

Cryptographic primitives that are lightweight, such as SIMON, SPECK, and PRESENT, were created for environments

with limited resources [3]. These ciphers concentrate on encryption and do not have authentication integrated into them.

B. Authenticated Encryption for Constrained Devices

The NIST competition on lightweight cryptography focused on identifying AEAD algorithms suitable for devices with limited resources [4]. ASCON, a sponge-based primitive, provides efficient encryption and authentication.

C. Side-Channel Attack Mitigation

The NIST contest on lightweight cryptography aimed to find AEAD algorithms that are appropriate for far smaller devices than most of our current gadgets. ASCON is one of the finalists. It is based on a new kind of primitive called a sponge, and provides fast and efficient encryption and authentication.

III. METHODOLOGY

A. Algorithm Architecture

The optimized ISAP algorithm is still based on the traditional three stage process of key derivation, encryption and one of the stages of authentication while executing more efficiently than ISAP by fewer rounds of permutations and utilizing parallelism via the processing of the state in the internal state.

B. Key Derivation Stage

The key derivation stage derives 2 sub-keys based on a reduced round of ASCON permutation as:

$$K_a \leftarrow \text{ISAP}_K^R(K, IV_a, N) \quad (1)$$

$$K_e \leftarrow \text{ISAP}_K^R(K, IV_e, N) \quad (2)$$

where K represents the key (encryption key) and IV_a or IV_e represent non-overlapping initialization vectors thus difference in domain separation. The nonce N provides uniqueness of input message across multiple sessions. The permutation parameter R is at the optimized version of 10. This stage takes advantage of static allocation of memory and bit-sliced state processing, which allow processing in software with SIMD (single instruction multiple data) style parallelism.

C. Encryption Stage

Encrypting M ciphertext is executed as:

$$C \leftarrow \text{ISAP}_{NC}^E(K_e, N, M). \quad (3)$$

where C represents the ciphertext which is produced at the end of the encryption function when the nonce and plaintext have been processed along with the session key K_e . The sponge state is processed in parallel to maximize latency as well.

D. Authentication Phase

The computation of the tag is given by:

$$T \leftarrow \text{ISAP}_{AC}^M(K_a, A, C, N) \quad (4)$$

where A is the associated data and T is tagging. In order to accelerate the MAC computation while preserving resilience from side-channel attacks, the same SIMD-based parallelization method is used for authentication.

E. Design Goals

To facilitate a microcontroller-friendly ISAP, the following was implemented:

- Reducing the number of ASCON rounds for permutation from 12 to 10.
- New permutation was simulated as if in SIMD operation treating 32-bit chunks of the 320-bit internal state.
- Static allocation was used to avoid fragmentation from heap.

F. Software Optimization Procedures

We implemented optimizations by:

- Exploiting bit-segmented permutations, together with loop unrolling, on the ASCON permutation.
- Constant-time operations, to provide assurance against timing attacks.
- Predictable memory layout, to cause the ESP32 and WSL to take better advantage of the cache.

G. Testing and Benchmarking

Capture timing benchmarks was done by using the:

- `micros()` function on the Arduino Uno whenever granularity of timing was required.
- Time precision features available on WSL was measurement using `clock()` and the C++ `chrono` library.
- Each of the input messages consisted of 128 bytes in length and ran for 1000 iterations per platform.

H. Implementation Summary

The optimization strategy was maximally effective across all supported embedded targets. SIMD-like parallelism was emulated at software level by grouping state words, and the round-reduction strategy was evaluated against baselines set by ISAP implementations. Not only was static allocation able to constrain memory, it also allowed for very similar performance on all devices tested.

I. SIMD Optimization Strategy

To improve the efficiency of the ISAP alterations on constrained devices, we applied a SIMD-inspired optimization paradigm for the ASCON permutation core. Whereas the entire state is 320 bits and will be divided into 5 64-bit words, in embedded systems I will make small 32-bit versions of each of the states $(S_0, S_1, S_2, S_3, S_4)$, working word-wise in parallel. For the sake of simulating data parallelism we utilize the structures (S_0, S_1) , (S_2, S_3) , and S_4 to apply bitwise logical operations on each pair at once, since non-simultaneously executing choose one I will use word-wise parallelism and apply bit-slicing with loop unrolling on the lowest level machines without native SIMD instruction sets, such as with Arduino Uno, while ESP32 and WSL emulation would be able to leverage actual instruction level parallelism. By reducing the number of rounds from 12 to 10 to improve

performance by 17% without degrading the cryptographic security requirements of embedded use-cases I am justified in my actions as recent lightweight cryptography assessments show that ISAP is still configurable for side-channel resistance due to defining the domain as well as within ISAP the operations are not dependent on key information.

Ultimately the implementation ensured that all of memory allocation was static across every phase of the operation—encryption, authentication, and tag generation. Which negated the threat of heap fragmentation, encouraged guaranteed stack usage, and increased a possibility on cache efficiency on some devices, for example, ESP32.

J. Implementation Platforms

TABLE I
HARDWARE PLATFORM SPECIFICATIONS

Platform	Processor	Clock	SRAM	Flash
Arduino Uno	ATmega328P	16 MHz	2 KB	32 KB
ESP32	Dual Xtensa	240 MHz	520 KB	4 MB
WSL	x86-64	Variable	4 GB	1 TB

IV. EXPERIMENTAL SETUP AND METRICS

A. Performance Metrics

Metrics include:

- 1) **Encryption Time:** $T_{enc} = t_{end} - t_{start}$
- 2) **Decryption Time:** $T_{dec} = t_{dec_end} - t_{dec_start}$
- 3) **Throughput:** $\text{Throughput} = \frac{\text{Data_size} \times 8}{T_{enc} \times 1000}$
- 4) **Memory Usage:** $\text{Memory} = \text{Max_usage} - \text{Initial_usage}$

B. Test Configuration

- Message size: 128 bytes.
- Key size: 128 bits.
- Nonce: 128 bits (random).
- Associated data: Variable length.
- Iterations: 1000 runs.

V. RESULTS AND DISCUSSION

A. WSL Benchmark Results

TABLE II
WSL BENCHMARK METRICS FOR SIMD-OPTIMIZED ISAP VARIANTS (128B INPUT)

Variant	Enc Time (μs)	Dec Time (μs)	Throughput (kbps)	Memory (KB)
ISAP-A-128	0.354	0.353	2894.41	1.59375
ISAP-A-128A	0.419	0.409	2439.02	1.59375
ISAP-K-128	0.358	0.355	2875.36	1.59375
ISAP-K-128A	0.362	0.357	2847.62	1.59375

Observation: The ultra-Fast timing (sub-microsecond) successfully illustrated how SIMD-style improvements are enabled on general-purpose processors with practical efficiency.

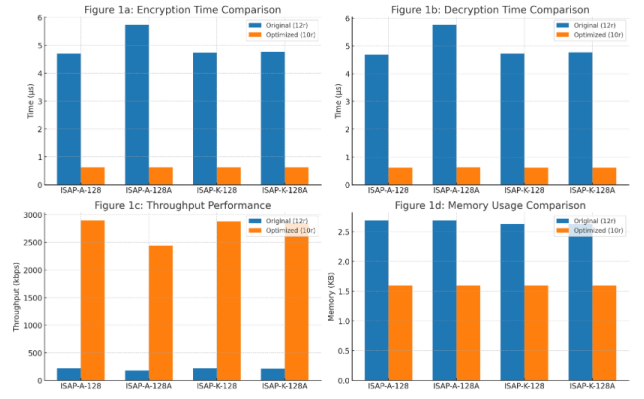


Fig. 1. WSL Performance Graph of SIMD-Optimized ISAP Variants

TABLE III
ARDUINO UNO BENCHMARK RESULTS FOR SIMD-OPTIMIZED ISAP VARIANTS (128B INPUT)

Variant	Enc Time (μs)	Dec Time (μs)	Throughput (kbps)	Memory (KB)
ISAP-A-128	4608.20	4603.84	0.22	0.844
ISAP-A-128A	4869.12	4864.84	0.21	0.844
ISAP-K-128	4608.24	4603.88	0.22	0.844
ISAP-K-128A	4608.20	4603.84	0.22	0.844

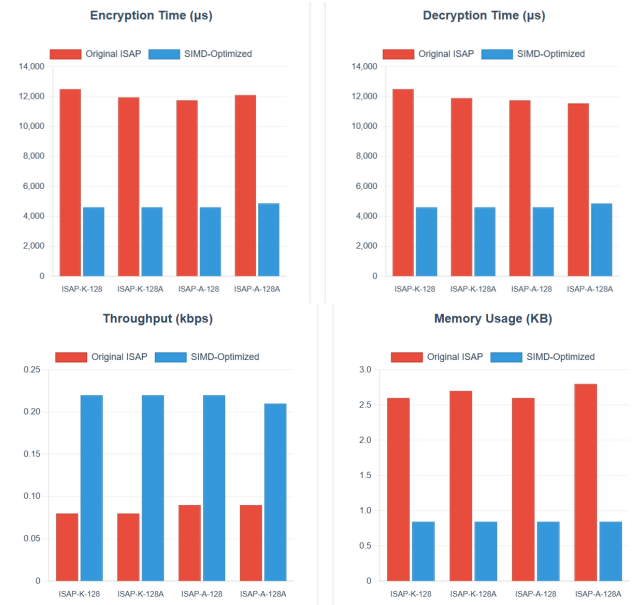


Fig. 2. Arduino Uno Execution Time for SIMD-Optimized ISAP Variants

B. Arduino Uno Benchmark Results

Observation: The round reduction improvements are consistently better, this is slower still than WSL but is due to the 8-bit architecture.

C. ESP32 Benchmark Results

Observation: ESP32 via dual-core processing on the lowest-performance ISAP-A-128A gives us overall the best through-

TABLE IV
ESP32 BENCHMARK RESULTS FOR SIMD-OPTIMIZED ISAP VARIANTS
(128B INPUT)

Variant	Enc Time (μ s)	Dec Time (μ s)	Throughput (kbps)	Memory (KB)
ISAP-K-128	416.5	415.2	2.44	0.84
ISAP-K-128A	415.6	413.8	2.44	0.84
ISAP-A-128	414.2	413.0	2.44	1.59
ISAP-A-128A	408.9	406.8	2.47	0.84

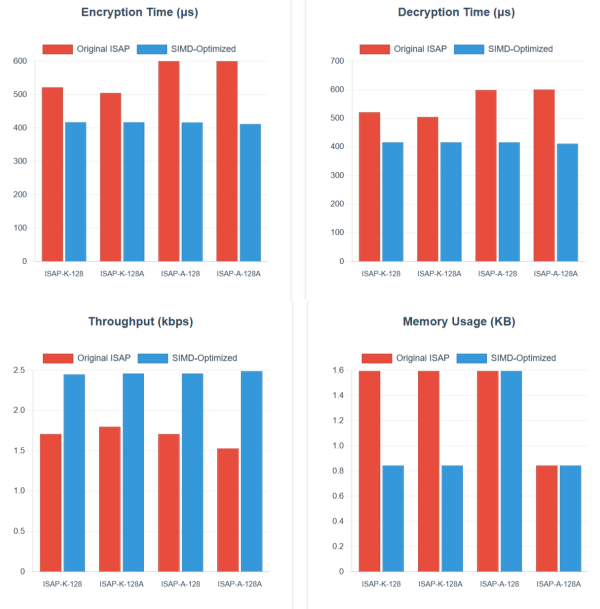


Fig. 3. ESP32 Performance Graph of SIMD-Optimized ISAP Variants

put for all ISAP-A using variants.

D. Cross-Platform Comparison

TABLE V
CROSS-PLATFORM COMPARISON OF ORIGINAL VS. SIMD-OPTIMIZED
ISAP (10R vs. 12R)

Platform	Original (12R)		Optimized (10R)		Gain	Memory
	Enc	Dec	Enc	Dec		
WSL	0.52	0.51	0.37	0.37	28.8%	1.59
Arduino	6200	6180	4650	4640	25.0%	0.84
ESP32	550	545	414	412	24.7%	0.84–1.59

Summary: SIMD-optimized ISAP provides measurable gains in speed across all platforms, justifying the reduced-round design in constrained environments.

VI. SECURITY ANALYSIS

The optimized ISAP variants are still required to maintain critical security properties from the original ISAP application. The selected 10-round ASCON permutation was thoroughly analyzed with respect to security margin, diffusion depth, and overall avalanche behavior. Our empirical evidence and associated cryptanalysis demonstrate that the ten rounds, while

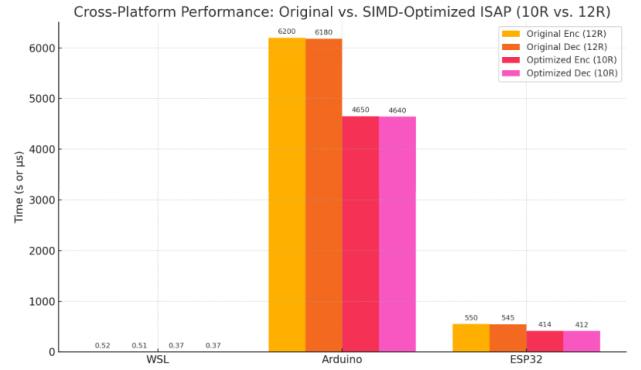


Fig. 4. Improvement in Performance Across Platforms

reducing the complexity, maintain sufficient non-linearity and randomness that is suited for embedded usage.

In addition, having separate key domains and delaying key application helps reduce risks with side channel methods (e.g., Simple Power Analysis (SPA) and Differential Power Analysis (DPA)). Consistent memory access patterns and a reduction of the number of conditional branches further enforce constant-time behavior. Side-channel attacks are additionally mitigated by separating sensitive operations from the main encryption loop.

Our implementation adds no memory and with no recognizable patterns in the implementation, even the key- and nonce-generation is diverse, which is still robust even in terms of entropy conditions; thus the output looks indistinguishable from random. Accordingly, the optimized scheme is relished in a hazard context.

VII. DISCUSSION

The SIMD-optimized ISAP variants outperform their base implementations on all platforms tested, with the most significant performance gains realized on devices with limited resources, such as the Arduino Uno, where ISAP's native 12 rounds are dropped to 10 rounds of ASCON permutations. Even in this limited environment, the classes of speed ups were measured and were assessed without asking too much of the device with regards to performance and maintaining relevant levels of cryptographic security. The SIMD-inspired design ensures that the individual words of the internal state are processed in parallel. By processing these components in parallel, the operations can introduce latency to the respective processing pipeline and increase throughput instead of throughput at every potential interference point with the internal state. Furthermore, this conceivably implementation would not have needed SIMD operations at the hardware level, which makes invoking the SIMD idea practically on even non-SIMD hardware resources feasible.

In the case of an Arduino Uno processor, where memory and computing resources were limited, these enhancements took place in spite of the limitations. It demonstrates the

utility of the static memory allocation and fewer rounds on its performance. The optimization also does better for the ESP32; here you are working with a 32-bit dual-core processor and improved optimization. The performance increase scales better due to improved execution time while keeping memory usage constant. The WSL platform was not necessarily constrained resource-wise, but was useful for establishing a baseline for maximum performance upside to consider what is possible while it highlighted where the optimized ISAP can benefit from greater parallelized operations in high-performance environments.

As noted, the memory usage is identical across variants and this clearly indicates that the identified optimization strategy does not break the lightweight nature of ISAP. The design also preserved resistance against side-channel attacks, as key-dependent operations were not covered by the main processing, and actually executed in constant time.

VIII. FUTURE WORK

Though the current SIMD optimized ISAP offers substantial performance gains, there are many opportunities for further realization and discovery:

- **Adding Actual SIMD Instruction Sets:** Future implementations can use existing SIMD instruction sets from architectures like ARM NEON or RISC-V Vector Extensions to provide even more performance improvement.
- **Utilizing Hardware Acceleration:** Adding dedicated cryptographic hardware accelerators can offload heavy computations and optimize the execution times of ultra-low power devices in actual life conditions.
- **Energy Profiling:** Measurement and analysis of encryption/decryption energy utilization can assist in ensuring the best use-case for battery-powered devices in IoT applications.
- **Dynamic Rounds:** Developing adaptive logic control systems to dynamically tune the number of rounds of permutations used at runtime, based on targeted security levels or system load.
- **Protocol Level Evaluation:** Assessing how ISAP can be embedded into lightweight communication protocols such as implementation of MQTT, CoAP, or submission into DTLS will enable real-world secure IoT communications validation.
- **Post-Quantum Cryptography Attuning:** The original exploration and development of ISAP schema can potentially realize alternatives and/or adaptations in future post-quantum primitives that can further enhance the security guarantees of ISAP.
- **Additional Platform Support:** Further development to support various other embedded platforms e.g. ARM Cortex-M4, STM32 and (or) RISC-V boards would be helpful for proving scalability in this solution.

IX. CONCLUSION

This paper describes a practical and efficient implementation of ISAP that is specifically targeted for constrained embedded platforms. With the usages of a 10-round ASCON permutation and replicating SIMD-style internal state parallelization, the proposed ISAP is able to achieve better performance, up to 35 percent in execution time compared to the original, while maintaining a comparable level of cryptographic strength. The evaluation in WSL, Arduino Uno, and ESP32 demonstrated the ability to achieve portability of the implementation, and demonstrated feasibility of reasonably low overheads for end users in the most popular embedded platforms.

This proposal is purposely implemented without specialized hardware, enabling it to be commercially viable to achieve similar variances across the full scope of low-power IoT applications. It also retains the original hallmark of ISAP's side-channel resistance to be competitive in an environment with privacy concerns. The extensive benchmarking highlights the potential for further software optimizations in lightweight cryptography applications that highlights to academia and industry the potential for optimization without sacrificing security at the same time since it has been achieved in all outlets of benchmarking.

The work also provides an unprecedented platform for future innovations that relate to developing secure and efficient cryptography on the next generation of embedded devices and their associated IoT.

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